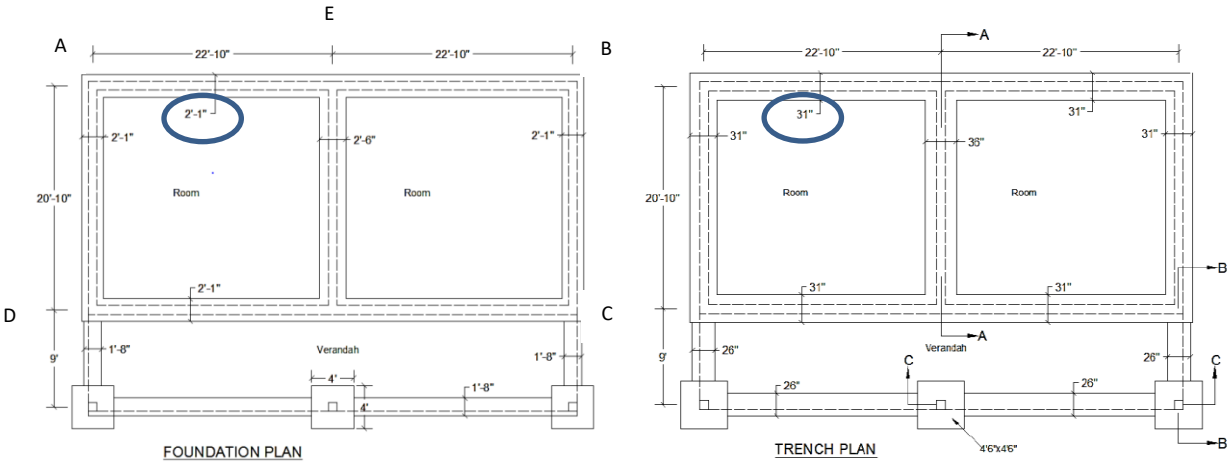
CE 208: Building Estimation Sample Calculations

1(ii) Excavation for 25" wall foundation

Length: ((22+(10/12)+(12.5/12)+(3/12))*4)+((20+(10/12)-(25/12)- (6/12))*2)
Width: (25 + 3 + 3)/12
Height: 3 + (3/12)

For excavation dimensions, we increase the actual width of foundation by 3 inch on all sides (Compare the foundation plan and trench plan). If trench plan is given, you can obtain the dimensions directly from trench plan.



The calculation shown above is based on the foundation plan. AB and CD is taken as continuous whereas AD and BC are considered to be in between. (But you can take the opposite too)

Along AE, Length of excavation = c/c distance + Half wall width of AD + Extra excavation

$$= 22 + (10/12) + (12.5/12) + (3/12)$$

(From trench plan, 22 + (10/12) + (31/12))

Along AD

Length = c/c distance – (Half wall width of AB and DC) - (Extra excavated portion of AB and DC)

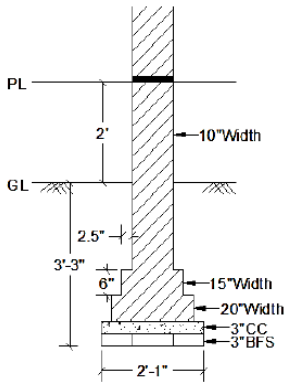
*Dimensions are subtracted as AB and DC have already been excavated

$= 22 + (10/12) – (12.5/12_{AB} + 12.5/12_{DC}) – (3/12_{AB} + 3/12_{DC})$

(From trench plan, 22 + (10/12) – (31/12))

Therefore total length: ((22+(10/12)+(12.5/12)+(3/12))*4)+((20+(10/12)-(25/12)- (6/12))*2)

Note: Dimensions used in calculation are in feet



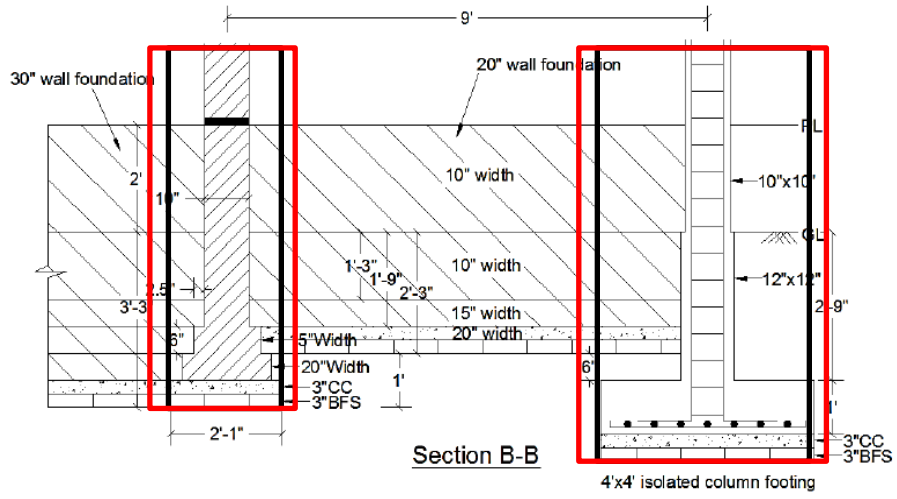
Width = $2 + (1/12) + (3/12_{Left}) + (3/12_{Right})$

Height below GL = $3 + (3/12)$

1 (iii) Excavation for 20 inch wall foundation

Length: ((9 - 2 - (3/12) - (12.5/12) - (3/12))*2) + ((22 + (10/12) - 4 - (6/12))*2)

Note: Since the depth of column footing is greater than 20" wall foundation, the former is excavated earlier.



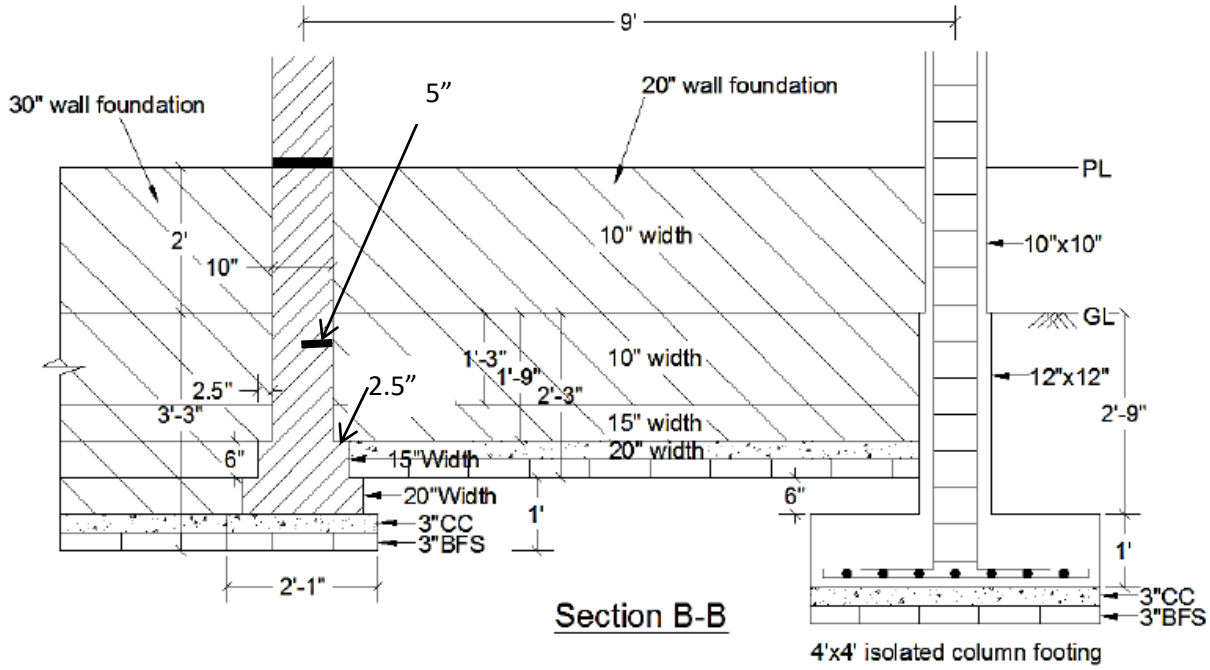
The black rectangles indicate the excavated portion if the actual width of foundation was used and the red lines indicate 3 inch portion extra excavated length on each side of foundation.

2. For brick flat soling (BFS) below foundation, we only calculate the area in order to determine the number of bricks required for covering the entire area. Cement concrete layer above BFS has the same width and length as in BFS. Only height is incoporated to determine its volume.

Note: Dimensions used in calculation are in feet

2(iii) Brick Flat Soling (BFS)

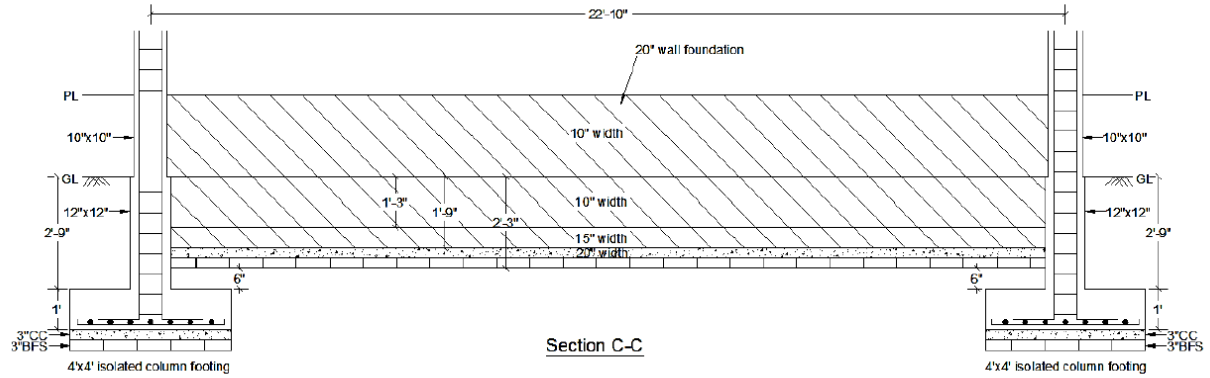
In order to find the length of **BFS**, you may consider the section view to find the relative height of foundations and obtain the lengths from the foundation plan.



9' is the center to center distance. BFS of 20" wall foundation meets the 25" wall foundation and the column footing at a depth of 2'3" below GL. At that depth, width of brick wall footing is 15" and column width is 12". So, from 9 feet you need to subtract half the wall width that is 7.5" (5" of inner portion and 2.5" extended portion) and half the column footing width that is 6"

Length: ((9 - 0.5 - (5/12)- (2.5/12))*2) + ((22 + (10/12) - 1)*2)

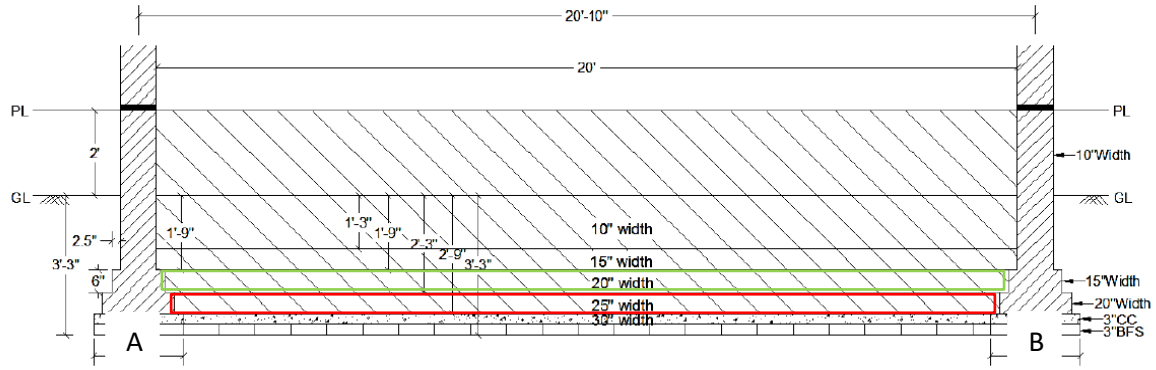
Width: (20/12)



Note: Dimensions used in calculation are in feet

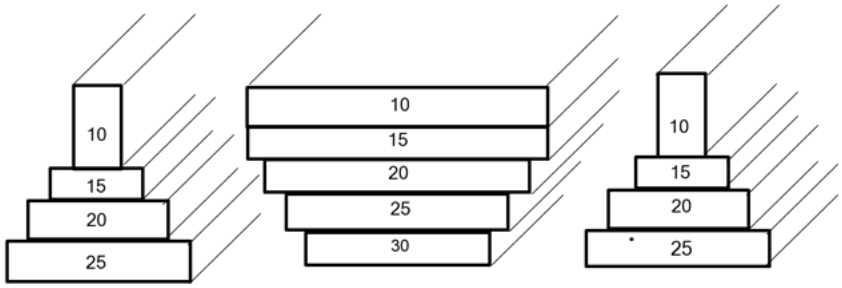
4. Brickwork in Foundation upto GL

(a) The 30 inch wall foundation has a series of steps. At the bottom, its width is 30 inch and gradually reduces to 10 inch decreasing by 5 inch in each step. The length along AB for each foundation width however increases by 5 inch as we move upwards.



Section A-A

The green block indicates the length of 20 inch width and the red block for 25 inch width.



For 10 inch width, the length can't increase by 5 inch since the width of 25 inch wall foundation is constant at that depth. The same length as for 15 inch width applies for the 10 inch width.

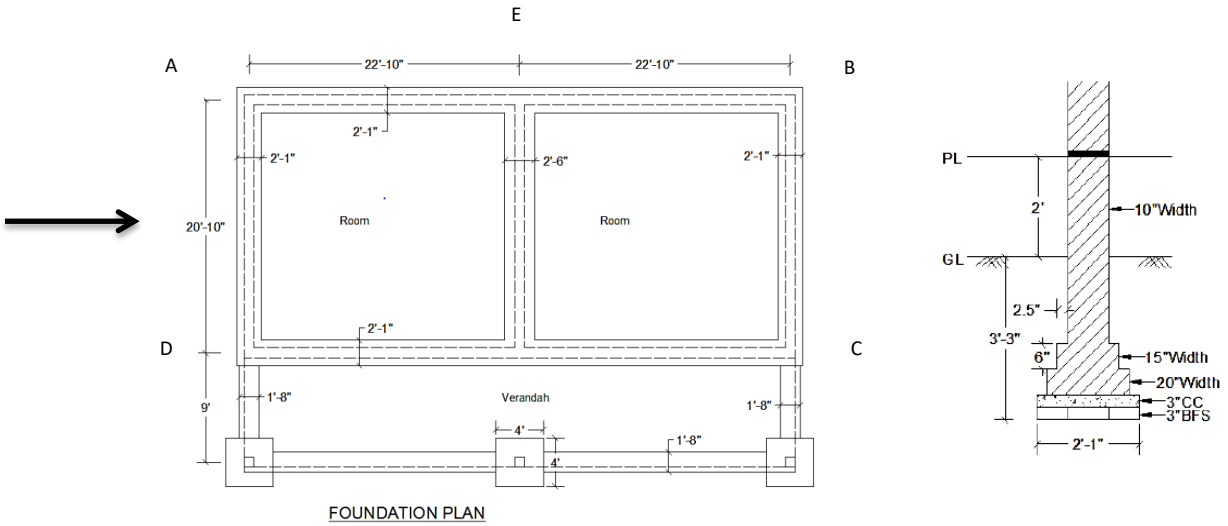
(i) 20 + (10/12) - (25/12) + (5/12)
(ii) 20 + (10/12) - (25/12) + (10/12)
(iii) 20 + (10/12) - (25/12) + (15/12)
(iv) 20 + (10/12) - (25/12) + (15/12)

Note: Dimensions used in calculation are in feet

4(b) 25 inch wall foundation

(i) 20 inch width –

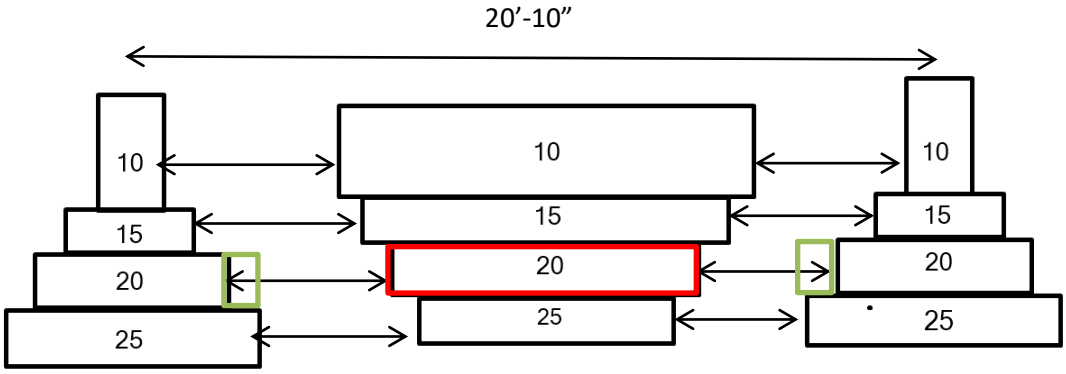
Length = ((22 + (10/12) + (10/12))*4) + ((20 + (10/12) - (25/12) + (5/12))*2)



The length of BFS for 25 inch wall foundation is as follows:

((22+(10/12)+(12.5/12))*4)+((20+(10/12)-(25/12))*2)

For the 20 inch width along AE, there is a 2.5 inch decrease in length at point A due to a reduced step size and hence ((22 + (10/12) + (10/12)). For the portion along AD, a side view (in the direction indicated by the arrow above) is shown below.



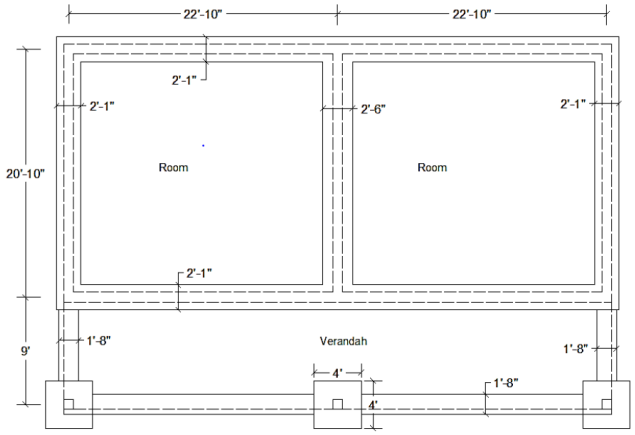
Therefore the required length = c/c distance – (2*Half width of 25 inch wall foundation) + (2*2.5 inch due to reduced step size) and hence ((20 + (10/12) - (25/12) + (5/12))

Note: Dimensions used in calculation are in feet

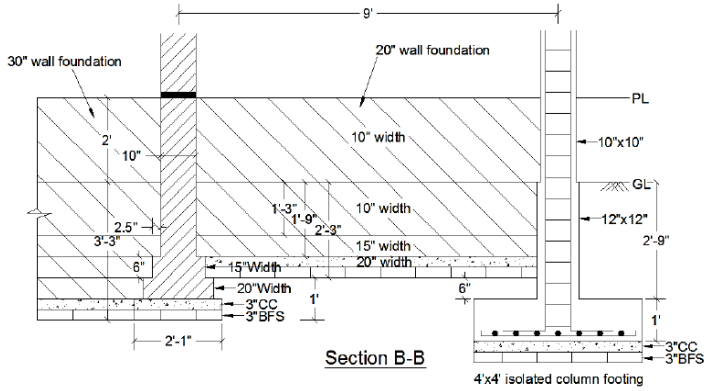
4 (c) 20 inch verandah wall foundation

(ii) **10 inch width** - ((9 - 0.5 - (5/12))*2) + ((22 + (10/12) - 1)*2)

You can use the following views to understand the sample calculation. Please keep in mind the elevation where the foundations intercept each other and calculate length accordingly.

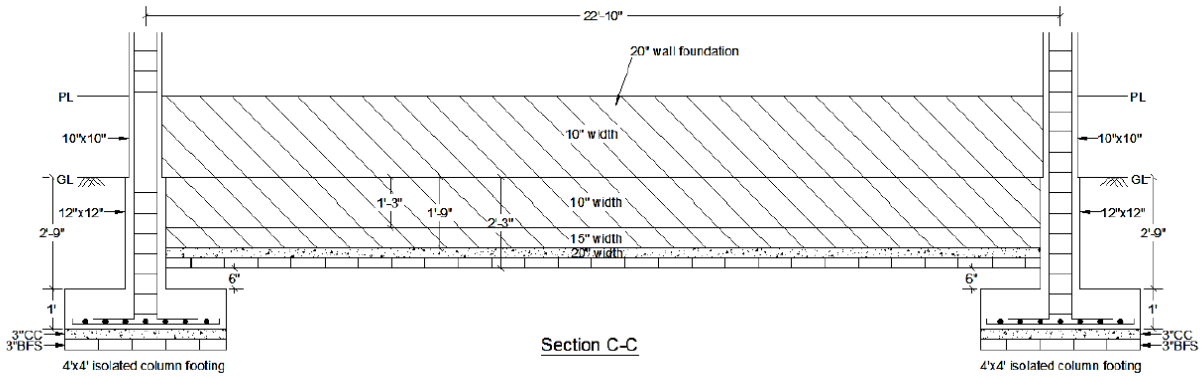


FOUNDATION PLAN



Section B-B

4'x4' isolated column footing



Section C-C

4'x4' isolated column footing

4'x4' isolated column footing

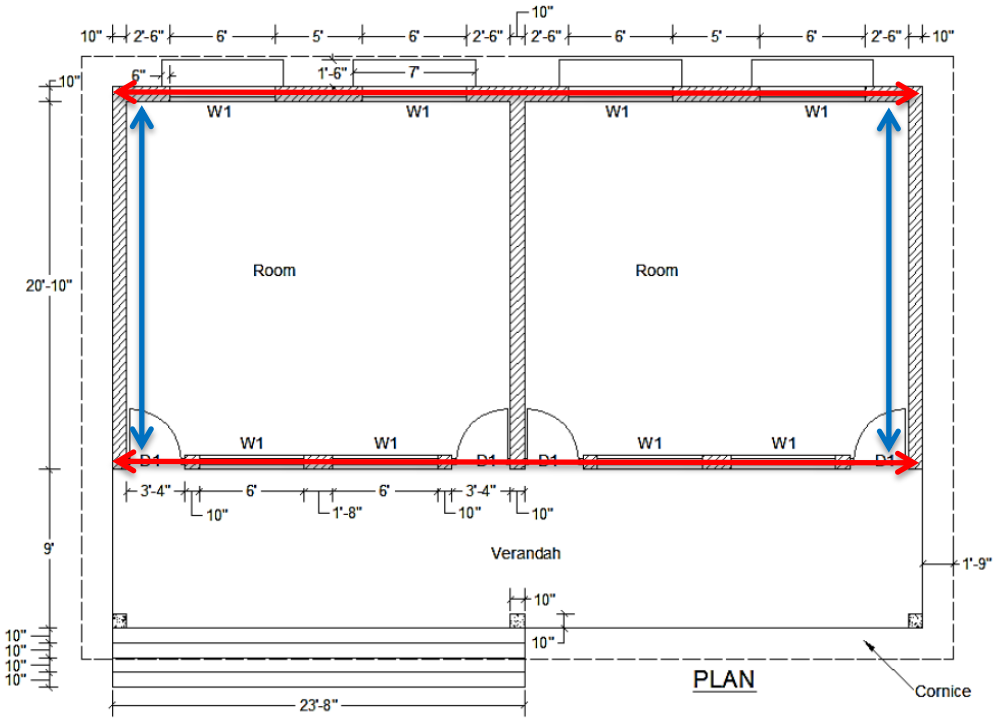
Note: Dimensions used in calculation are in feet

5. Brickwork in Foundation from GL to PL

(ii) 25 inch wall foundation –

Length = (22+ (10/12) + 22 + (10/12) + (10/12))*2 + 20 + 20

Use the plan for calculation of lengths as wall thickness is constant at 10 inch throughout. The dimensions are shown by arrows.



Note: Dimensions used in calculation are in feet